

UX Research Data Gathering

Highlights

- The concepts of work, work activity, work practice, and work domain
- Cast a broad net to understand work practice and the market
- Before the visit: Prepare to meet with customers and potential users
 - Learn about client, subject-matter domain, proposed product/system
 - Choose visit parameters, goals
 - Organize data gathering team
 - Recruit user participants
- During the visit: How to collect UX research data
 - Observe and interview clients and users
 - Kinds of data to look for
 - How to capture work activity data with effective raw notes
- Roots of UX research

6.1 INTRODUCTION

6.1.1 You Are Here

We begin each process chapter with a “You are here” picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 6.1). Within the understand-needs lifecycle activity, this chapter is about the data gathering sub-activity, in which you observe and interview the client and users to acquire a thorough understanding of user work¹ practice using an existing product or system.

Data gathering, in this chapter, is an empirical process to seek, extract, and gather real user work activity data. Data analysis in [Chapter 7](#) is an inductive (bottom-up) process to organize, consolidate, and interpret the user work activity data. [Chapters 8](#) and [9](#) are about the synthesis of various design-informing models (e.g., task descriptions, scenarios, user personas),

¹The term *work* broadly signifies anything people/users do while using products or systems (see Section 6.2).

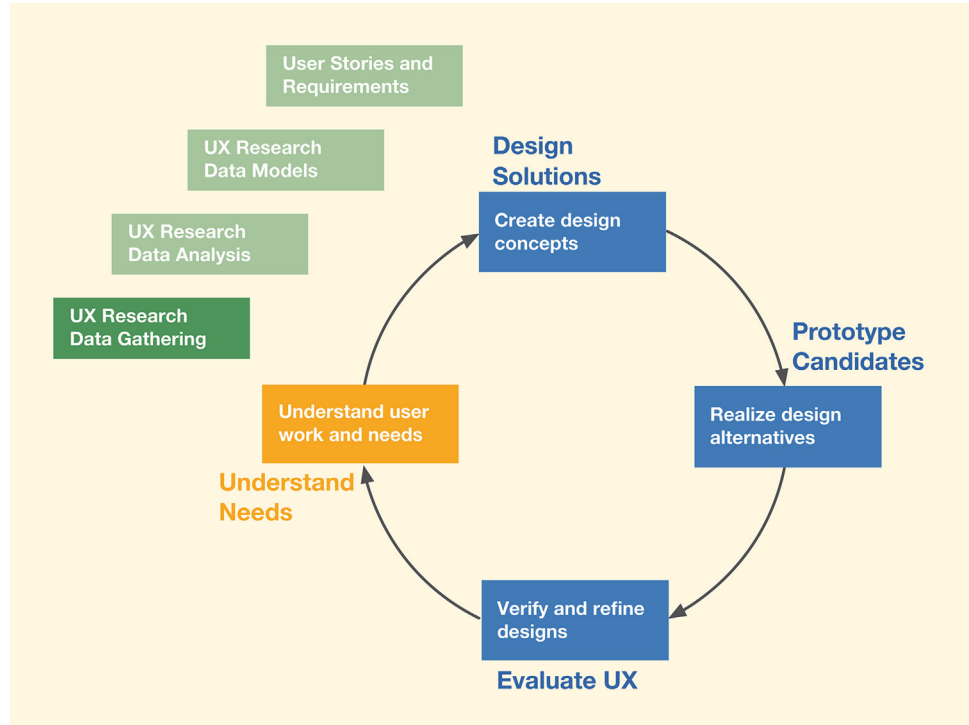


Fig. 6.1

You are here in the UX research data gathering chapter within the understand-needs activity, in the context of the overall Wheel UX lifecycle.

and [Chapter 10](#) is about a deductive analytic process for organizing and representing user stories, needs, and requirements.

6.1.2 Are We Studying an Existing Product/System/Work Practice or a New (Envisioned) One?

This is a question we often get from students when discussing user research. The essence of the answer is that you are not really studying a product or a system; you are studying work practice in the context of using a specific product or system. At the end of the day, the answer might be “both existing one and new one.” Researchers, designers, and users can be biased toward thinking ahead to the new (envisioned) system and even a new work practice, but almost everything we do in UX research begins with the existing system and work practice. For almost any new product or system, there is almost always some kind of existing practice. For example, there may not have always been an Apple iPod, but there have been portable music players for a very long time.

6.1.3 About User Research

User research is the most popular term for the process of getting to know the users, usage, user work, UX, work context, and UX design needs in preparation for UX design; however, this term fails to tell the whole story. Yes,

we definitely want to know the user, but it is also essential to study the deeper nature of the work users do and the more complete setting of the users and work that we call the domain or work domain (referring to user work while using the existing system or the one being designed). As such, we will use UX research, which is a broader term than user research.

6.2 TERMINOLOGY RELATED TO USAGE OF PRODUCTS AND SYSTEMS

An important goal of UX research is to understand the work people do with a system being studied and how they do that work. To that end, we need to establish a framework for discussion. We begin by introducing a few related terms.

6.2.1 The Quest for Broader Terminology

Many discussions of human activities to accomplish a goal are based on so-called work, as seen in the literature on work activity theory (Bjerknes, Ehn, & Kyng, 1987; Bødker, 1991; Ehn, 1988). However, using work does not allow discussion of other ways people use products and systems. For example, it does not accurately describe people's usage of products or systems for fun, play, or other personal goals. A more general term—(activity)—came to mind because, like work, it is something users do, but it does not play well with the rest of the terminology we need here.

At the end of the day, we decided to see it through with the original simple term (work) for these reasons:

- There is already a body of literature based on this term.
- The prominence of this term in our discussion dictated simplicity, economy, and ease of use.

6.2.2 Work

Therefore, we use work to refer to what needs to be done using products or systems within a given problem domain, often with a high-level personal or business goal. In this context, we ask the reader to please accept it to signify a broader concept that:

- Embraces all things users do, including activities beyond work and play
- Includes a broad range of human activities to which UX design applies
- Implies that these activities are usage activities, in the sense that they are activities involving the use of a product, system, device, or other technology
- Excludes simple activities, such as taking a walk, for which technology is not a focus

6.2.3 Work Activity

Work is a good term for the general concept; however, work activity refers to a specific instance of work, a broad set of related things a person would do as part of one goal. Under this broader interpretation of work, examples of work activities span a vast number of areas, including designing a car dashboard, entering data into a form, playing a video game, voting in an election, driving a car, reading email, listening to music, using a coffee machine, using an automatic teller machine, selling tickets for entertainment, reading signs to find your way in a hospital, and receiving healthcare in a clinic. A more specific example of a work activity is using a CAD/CAM application to design an automobile. Job-related activities such as this usually have a high-level business goal, which in this example may be to lead the family sedan segment of the automobile market.

6.2.4 Work Practice

Work practice is a complete description of how all related work activities happen (or should happen) in a particular context. That context could be the activities of just one person, or it could be reckoned within an entire organization. More specifically, work practice is a description of how people make use of a product (such as a consumer product, personal device, or software product) or system (such as a large enterprise system) to get an activity done.

If the context of a work practice is an individual person or a single user role, then the work practice description will include activities and characteristics of just that person and the context of the technology ecosystem. For example, the work practice of a person who listens to music on an iPhone may include the importance of music to that person and preferences for types of music alongside the context imposed by Apple's Music application, its subscription details, and other governing policies. Furthermore, the listening activity may intertwine with other activities, such as listening to music while exercising, which, in turn, may extend the work practice to include the Apple fitness app and corresponding subscription details.

In an organizational context, work practice would include a description of all work roles—that is, all activities of those roles, procedures, traditions, customs, and protocols associated with doing these work activities, usually as a result of that organization's goals, user skills, knowledge, and social interaction on the job. A concrete example of such a work practice is a complete description of how the automobile portion of the Ford Motor Company performs its activities. This work practice is likely different than

the work practice at Tesla, for example, because a single work practice is influenced by factors operational within a single organization, including the users, organizational culture, and what the branding and values say about the organization.

6.2.5 Work Domain

Work domain is the broader milieu of an entire industry, such as healthcare, cloud technology, or finance. Work domain is an abstract, all-encompassing, and expansive description of the nature of work across all organizations in a particular area of interest or endeavor, including work roles, systems, regulations, technologies, norms, first principles, conventions, and variations among work practices across that area. To put it simply, the work domain of X is the world of X. An example of a work domain is the manufacturing of consumer automobiles. So, the work domain for car manufacturers would include the work practices across all car manufacturers, including Ford Motor Co. and Tesla, for example. Ford and Tesla will probably have quite different organizational work practices, but both fit within a broader common work domain of car manufacturing. An example that goes beyond the work practices of separate organizations is the common connections to skills and traditions of the craft of the domain.

6.2.6 Work Role

A work role is not a person, but a position assumed by a human user. Such a role is defined by the expected activities and behavior of people in that role within the context of a specific work practice. If that role is job related, for example, it is defined by the duties, functions, and work activities of a person with a certain job title, or job responsibility, such as customer or database administrator. In the auto manufacturing world, examples of work roles include automobile designer, assembly line foreman, and product marketing lead. In a more casual or personal realm, an example of a work role could include a user engaging in work activities, such as organizing, curating, listening to, or otherwise managing a music collection.

Because a work role is defined in terms of activities performed by the role, it is an agentic kind of entity, an entity that performs. The agentic view answers the question, What is the reason for this role? This is an important attribute for UX research to unpack and understand. The reason for a role directly ties to the goals of the business at hand and acts as the charter for what the design needs to support in the product or system.

6.3 WORK PRACTICES WITHIN A WORK DOMAIN

In the course of doing UX research, we want to learn about the relevant work domain and one or more work practices within the domain (Fig. 6.2). This often includes:

- A. The details of each work practice: Who are the users, and what roles do they play in a given work practice? What are the goals, motivations, challenges, and preferences of these goals? What are their work activities and tasks?
- B. Relationships among the work practices: How does work in one work practice contrast with that of other work practices? This includes competition, differentiations, cultural norms, branding, and values.
- C. The work domain itself: The general nature of the work domain, externally imposed characteristics shared over all work practices, including regulations, market trends, and technology trends. Plus, this includes internal characteristics of the work, such as its first principles and concepts.

6.4 GATHERING DATA FOR UNDERSTANDING WORK DOMAINS AND PRACTICES

Much of what we do in UX research develops the picture of the work domain and work practices at the same time. The kind of data you, the UX practitioner, want to collect and the ways to obtain it will depend on your goals.

If your UX research is based only on talking with users in one organization, you are only looking at a single work practice (e.g., A in Fig. 6.2), which does

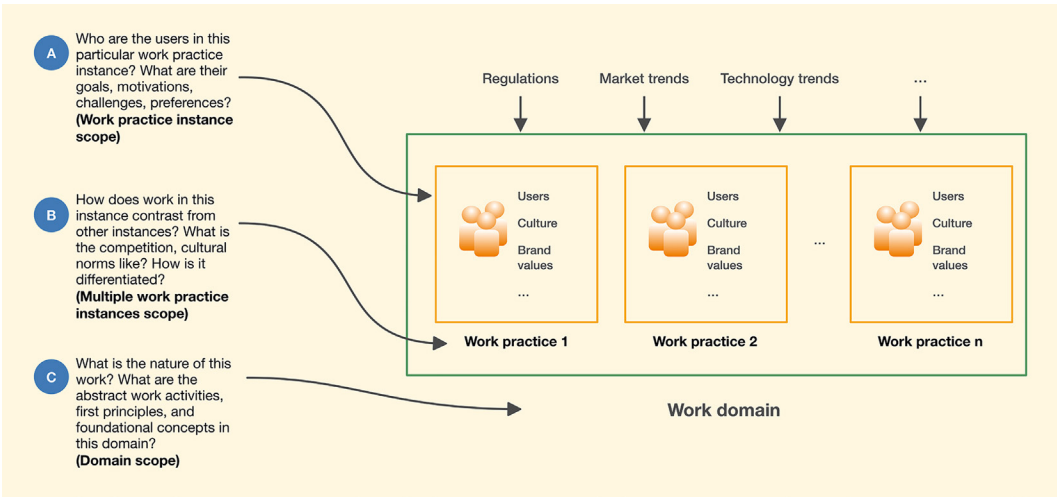


Fig. 6.2
The relationship between a work domain and the work practices it contains.

not help you achieve a depth of understanding about the whole domain (e.g., C in [Fig. 6.2](#)). The external factors on the outside of the domain box at the top of [Fig. 6.2](#) (regulations, market trends, etc.) and the common internal characteristics inside are attributes of the domain and common to all work practices. Learning about and understanding a work domain outside a particular work practice relies on tapping into whatever resources are available.

Learn about the work domain. Do your homework and internalize the work domain of the product or system (e.g., C in [Fig. 6.2](#)). Talk first with the client and subject-matter experts (SMEs), and then validate this understanding with users:

- Understand the concepts and first principles in the domain (e.g., for a payroll system, how pay is computed for a pay period, how withholdings work, how to withhold taxes and the basis of their requirements, what reports to submit to the government, and differences among the various forms of compensation).
- Learn about the culture of the work domain in general (e.g., understand the precise and conservative financial domain versus the laid-back nature of the art domain).
- Know and understand the vocabulary, technical terms, acronyms, or slang of the product or system work domain.

Learn about the client company/organization work practice. Explore your client organization's work practice (e.g., A in [Fig. 6.2](#)) to learn about:

- Business goals
- Organizational policies and ethos (scroll the organization's online and social media presence)
- Any existing legacy systems (outdated systems with maintenance problems that date back possibly many years)
- The product or system proposed for your work practice. Check the initial product or system conceptualization documents (see [Section 5.3](#)) for early statements about system functionality and structure and even the system architecture. Try out the company's existing and previous products and read reviews.
- Customer sentiment about products and market reputation

Learn about relationships of your organization's work practice with other work practices in the domain. Look at how your work practice compares with others in the work domain (e.g., B in [Fig. 6.2](#)) in these ways:

- The competition
- Shared goals and challenges

- Business advantages in your favor
- Products and systems used within other work practices

Proposed system may be internal to a single organization. If your client is an organization seeking to develop a product or system for internal use by that organization, the scope of your work will be mostly limited to that work practice. By interviewing and observing users at work, you will get many perspectives on the same work practice, and the data will primarily be about one work practice (e.g., A in [Fig. 6.2](#)).

You may also benefit from looking at work practices of other organizations (data from B in [Fig. 6.2](#)), especially to see if you are missing something in the view of the world represented by your work practice. Explore a few other work practices if you have access and the resources to do it.

Developing commercial tools for the domain. If you are developing tools to support work practice across a work domain, your UX research must consider a broad range of possible work practices (data from B and C in [Fig. 6.2](#)).

6.5 THE NATURE OF UX RESEARCH DATA GATHERING

The overarching goal of UX research is to understand the work domain or a specific work practice through the lens of existing users by visiting their work context and learning about them, how they do work, and what their work practice entails, including the challenges or hurdles they face and any workarounds they employ.

Because learning about work practice is a large part of UX research, the rest of this chapter is mostly about observing and interviewing people—users and others familiar with the work practice. Here, we seek data in A of [Fig. 6.2](#), and we describe data collection as a process.

Data gathering is a prelude for requirements. UX research data gathering is not yet about requirements, but it is essential for understanding how users do their work. This will lead to insights about user needs and what it would take in a system design to support them.

Not the same as task analysis or a marketing survey. Your client may say, “We already do that. We do task analysis and marketing surveys.” How should you respond to that? The simple answer is that they are not the same.

Task analysis is a systematic human factor technique of examining user tasks by studying the structure of tasks and task steps. This does not give enough insight into situations where tasks are interwoven or where users need to move seamlessly from one task to another within the work context. This

also does not tell us about user motivations, friction areas, breakdowns, and overheads associated with users performing closely related tasks using multiple systems.

Similarly, you cannot substitute market research for UX research. Marketing data are about sales and can identify the kinds of products and even features customers want, but do not lead to enough understanding about how people work or how to design for them. They are just two different kinds of analysis, and you may need both.

UX research is not about asking users what they want. Rather, UX research is about understanding user work practice and work activities in that context. Your job as a UX team member is to deduce their design needs (see [Chapter 10](#)).

Data gathering requires real detective work. It takes determination to learn about other people's work. Going beyond how users are supposed to do the work or even how they say they do it requires digging into the details outside the idealized practice. People creatively solve and work around problems, making barriers and problems less visible to them and to outsiders studying the work. Tease out clues about work practice to construct a complete picture of the work, details of which can be hidden beneath the surface observables, including intentions, strategies, motivations, and policies.

Example: Discovering the Area of Inventory by Following Leads

As an example of following leads, this is a real story told by a team doing a project for one of our classes. The client was in retail sales, and the conversation part of the interview centered on that concept, including serving customers, making the sale transaction, and recording it. During this conversation, however, the client mentioned inventory once in the context of point-of-sale data capture. We had not asked about inventory, so no one had mentioned it until this point.

Our good ethnographic detectives, recognizing a clue revealing another area of work activities, pounced on the topic of inventory and pursued a new train of thought. What about inventory? What role does it play in your point-of-sale data capture? Where does it go from there? How is it used and for what? How do you use inventory data to keep from running out of stock on items in demand? Who orders new stock and how? Once an order is sent, how do you keep track of it so it does not fall through the cracks? What happens when the new stock is delivered? How do you know when it arrives? Who works in receiving, and what do they do? How do you handle partial shipments? How do you handle returns?

Cast a broad net to include the work domain. Each time you gather UX research data, it provides a different viewpoint into the work practice. Remember, too, we are not just looking for user data; we also need to understand the work domain. It is up to the UX researchers to put the pieces together and assemble the larger overall picture.

Understand the work practice ecology. It is important to understand the broader connections among the work activities that are perhaps not the direct focus of the design team. Examples include outside systems that users of the target system rely on to get their job done. For example, do they use external news sources, data feeds from other services, or third-party payroll systems? How do those systems interface with their immediate system in getting work done?

Understand how information is organized, the artifacts that represent views of it, and how work flows. Map out information architecture and key workflows in current work practice. For enterprise systems, ask for screenshots of key screens and samples of reports or other work.

Understand market forces and trends. To be truly innovative, the UX team must understand the client's market perspective, what the competition is doing, the broader trends, and the leading players.

Your sources must be protected. Except for unusual circumstances, you are ethically required to maintain confidentiality and anonymity of anyone you observe or interview throughout your entire UX research process. Make sure to establish, with the client, the need for freedom to comment honestly and without undue political and managerial influences. This is especially important in cases where you have observed problems and breakdowns arising due to social and political issues in the work practice.

Talking about situations involving breakdowns due to bad management or flawed work practices (modeled in social models) is especially dangerous for your participants if there is a chance the sources will be revealed. Make this your unbreakable rule: When you take data and models back to anyone, users or management, all participants must remain anonymous. Otherwise, they will be reticent about coming forward with valuable insights.

6.6 BEFORE THE VISIT: PREPARE FOR DATA GATHERING

Your goal for data gathering is to learn everything you can about the client, the company, the business, the domain, the product or system, and the users. In this way, you are prepared and productive during field visits.

6.6.1 Decide on Your Data Sources

The first choice of a data source for understanding user work practice and needs is often users who actually use the current product or system in the field. Our first choice of data sources for understanding the work domain is a competent subject-matter expert (SME).

Supplemental or alternative data sources include:

- Dual experts
- Focus groups
- User surveys
- Competitive analysis
- Domain expertise through education
- Being your own domain expert

Interview SMEs. Subject-matter experts have a deep understanding of a specific work domain and the range of work practices within that domain. First interviewing SMEs is a fast way to establish the essence of a domain. Even though users can probably give you the best user data information (e.g., unpredictable problems that crop up in real usage), SMEs can give other important kinds of information, such as an inside view of how a system should work. They can also provide deeper insights into variations of the work practice across that domain.

Use dual experts. Sometimes you are fortunate enough to recruit dual experts (i.e., experts in UX and the work domain). An example is a designer of Adobe Lightroom (a high-functionality software application for managing and processing large collections of images and photographs) who is also deeply involved in photography as a personal hobby.

Listen to focus groups. A focus group (Krueger & Casey, 2008) is a technique wherein a small group of representative users or stakeholders discuss broad issues introduced by a moderator. A focus group can be useful in prompting a revealing conversation about more complex issues or themes pertaining to a work practice or domain.

Employ user surveys. User surveys are great for prioritizing approaches. You can ask specific questions about the work domain or you can ask for open-ended feedback (e.g., aspects they like or dislike about a product or work practice).

Do competitive analysis. An analysis based on comparison with market competitors can expose strengths and weaknesses and can help identify capability gaps and shortcomings.

Acquire domain knowledge through education. Sometimes you can prepare for UX research by educating yourself about the core concepts, technologies, business practices, and trends in the work domain.

Be your own domain expert. Sometimes, if you are a user (especially of a consumer product or device), you can rely on your own experience and insights to understand users and their needs. For example, Apple designers are also Apple product users.

6.6.2 Choose Visit Parameters

Depending on the nature of your project, you need to decide as a team on the following:

- How many visits can or should you make?
- How many users can or should you interview per visit?
- What kind of user roles and users should you involve?

The answers are usually found in what it takes to meet your goals, subject to constraints of budget and schedule.

Less rigorous data gathering. When you need to be economical in your parameter choices, especially for agile UX environments, it is important to take the following advice:

- Interview and observe fewer users
 - Choose fewer numbers of more experienced users; you will learn to squeeze out a great deal of useful information from a small number of users.
- Use fewer observation and interview sessions per user
 - Just one day's worth of talking to users about their work practice can make a big difference in your understanding of the work domain to inform the design.
- Be smart in capturing data
 - Abstract your notes down to the clean and uncluttered essence. As you gain experience, you will excel at filtering inputs on the fly.

More rigorous data gathering. Very large enterprise systems, such as a new air traffic control system, are outliers and will not be encountered by most UX professionals. However, if you are working on such a project, a high level of rigor may be necessary, and you should understand the steps to achieve it. In such design situations, include users for all roles and experience levels in that work practice as well as experts or users from other work practices to understand variations in that domain.

6.6.3 Data Gathering Goals Based on Scope

In agile UX, you may end up doing UX research in both the large-scope early funnel (see [Section 4.4.4](#)) and the small-scope late funnel (see [Section 4.4.3](#)). This will usually entail two different kinds of UX research at different levels and occurring in two different parts of the overall UX lifecycle process.

Early funnel data gathering. The early funnel is the part of the agile UX funnel for large-scope activity, usually for conceptual design, before syncing with agile software engineering. Here, you will conduct high-level UX research to obtain a general overview to establish a high-level system structure and build up inputs to the broad conceptual design. You will focus on high-level work activities plus more general user stories, user needs, and requirements.

Late funnel data gathering. In the late funnel (the part of the agile UX funnel using small-scope activity for syncing with agile software engineering sprints), you will focus on one (or a few) specific user stories at a time to obtain detailed inputs to drive low-level interaction design for task sequencing and navigation. You will gather supplementary information to refine your models, answer questions, and fill in gaps.

6.6.4 Organize Your Data Gathering Team

Your goal in organizing a data gathering team is to pick people with the skills appropriate for this client and domain to be effective in, and capable of, understanding this work practice.

Often, one person can do the job, but two may be better. Set your own limits on the number of visits to make, depending on your information needs, budget, and schedule. A multidisciplinary mix is always best (e.g., user experience people, SMEs, product managers, and others familiar with the product domain).

6.6.5 Recruit User Participants

If you decide on users as your data-gathering information source, get help from your client in selecting and contacting a broad range of participants from the potential user population. Recruit locally for users of consumer products.

Plan to see multiple users for enterprise systems. Each user may have a different take on how the broader work domain functions. Include the following:

- All work roles (see [Section 8.2](#)) in that work practice
- Grand customers (customers of the customer, if any) outside the user's organization

- Indirect users who need direct users to interact for them (e.g., ticket buyers at a movie ticket window need ticket sellers to interact with the system)
- Managers

Make every effort to get access to important but so-called unavailable people. For projects in some domains, you may be told that users are busy and unavailable. For example, management might resist giving access to key people because bothering them would cost the organization time and money. Make the case for meeting at least some of these users to develop an understanding of their work activities and the potential cost of not including their input on the new design.

Get permission to observe users in their work context. The user context of interviews should be as close a match as possible to the usual working (or playing) location and conditions. Avoid being led to a conference room, for example, because they say it is quieter and less distracting there.

6.6.6 Prepare Your Initial Questions

Just as football quarterbacks sometimes script their initial series of plays, we recommend that you script your initial interview questions to get you off to a good start. Ask them to tell you and show you how they do their work. What tasks do they typically do? What actions do they take? With whom and with what do they interact? Ask them to demonstrate what they do and to narrate it with stories of what works, what doesn't work, how things can go wrong, and so on.

6.7 DURING THE VISIT: COLLECT USER DATA

6.7.1 Set the Stage Immediately

- Get off to a good start by establishing rapport with the client and each user.
- Explain the purpose of the visit: To learn about how people use, or will use, the product or system and to inform design.
- Explain your approach.
- Promise personal and corporate confidentiality.

6.7.2 Observing Versus Interviewing: What They Do Versus What They Say

Observation. Observation (a technique from the UX technique toolbox; see [Section 2.3.1.1](#)) can help you see work activity with an independent eye. It is not enough to depend on users trying to describe what they do, especially in work

that has been internalized. Humans are notoriously unreliable about this. Simonsen and Kensing (1997, p. 1) explain, “A major point in ethnographically inspired approaches is that work is a socially organized activity where the actual behavior differs from how it is described by those who do it.”

Example: Observation of a Car Wash Workflow

An example of the observation technique in a real-life setting is observing a car wash workflow problem. On a recent trip to Buster’s Auto Spa, our local car wash, I had to sit in line almost half an hour to get into the car wash. I guess most people would read the paper or listen to music to pass the time. Being blessed/cursed as a UX designer, however, I was curious about the cause of the delay.

[Fig. 6.3](#) shows a simplified sketch of the entrance, which is like most any car wash. Note that this is a practical example of at least two other techniques, abstraction (see [Section 2.3.3.2](#)) and sketching (see [Section 2.3.1.3](#)).

Most of the time the car wash works well, but occasionally it has a workflow problem. When a customer wants the interior of the car cleaned, the attendants use a large vacuum cleaner mounted in a fixed location at the entrance. Vacuuming can take 10 minutes, during which time no cars are moving through the car wash—bad for the car wash company and bad for customers who have to wait in line behind the car being vacuumed. Customer waiting is especially bad because getting your car washed is something you usually do on the way to something else. Further, the narrow lane leading up to the entrance makes it almost impossible for customers to bail. Of course, I began to think about different designs of the facility that would solve the problem. What would you recommend?

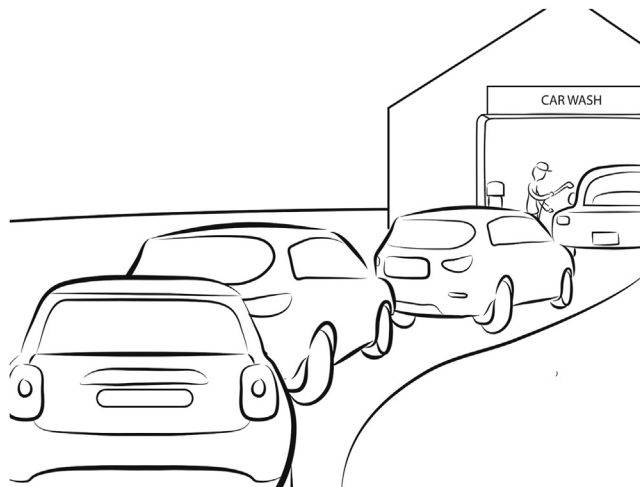


Fig. 6.3
Entrance to Buster’s
Auto Spa.

Exercise 6.1 Make Some Deeper Observations

Observe a customer service counter at Costco or similarly busy department store and prepare findings. Include descriptions of what is taking place, the flow, breakdowns, etc.

Interviewing. Using observation alone, you can miss some important points. For example, an important problem or issue simply might not come up during any given period of observation (Dearden & Wright, 1997). Remember to ask about corner cases and activities that are performed only at certain times (e.g., payroll at the end of the month) or are infrequent (e.g., when an external regulatory agency conducts an inspection). Narration by the user can augment your observations with information about motivation, feelings, special cases, and other hidden aspects. Get your participants to demonstrate all the ways they use the product or system and have them think aloud (Sections 22.3.2 and 22.3.3) as they work, revealing their motives, rationale, and perceptions of UX problems.

6.7.3 Shadowing and the User Journey

Data gathering can benefit from shadowing, a technique in which the UX person shadows a user, taking notes about typical usage within the daily routine (i.e., user journey).

Example: The Black+Decker SnakeLight—An Example of Shadowing and the Value of UX Research in General

In the mid-1980s, Black+Decker was considering a move into the flashlight market, but the company did not want to join the crowded field of ordinary consumer devices. To glean new ideas, designers shadowed real flashlight users and documented their customer journeys. They observed people using flashlights in the context of real usage situations and discovered the need for a feature that was never mentioned during the usual brainstorming among engineers and designers or in focus groups of customers. More than half the people they observed looking under car hoods, under kitchen sinks, and in closets and attics said that they needed both hands free to do the task. So the flashlight they designed allowed the user to change its shape, letting it stand by itself on a workbench or other surface, such as under the kitchen sink. Fig. 6.4 shows a Black+Decker SnakeLight in the context of usage. It is very flexible and even looks a little bit like a snake.

Without benefit of this UX research, Black+Decker designers would have had to rely on a purely diagnostic view. They would have looked at a typical flashlight and fixed all of the problems they could find with that

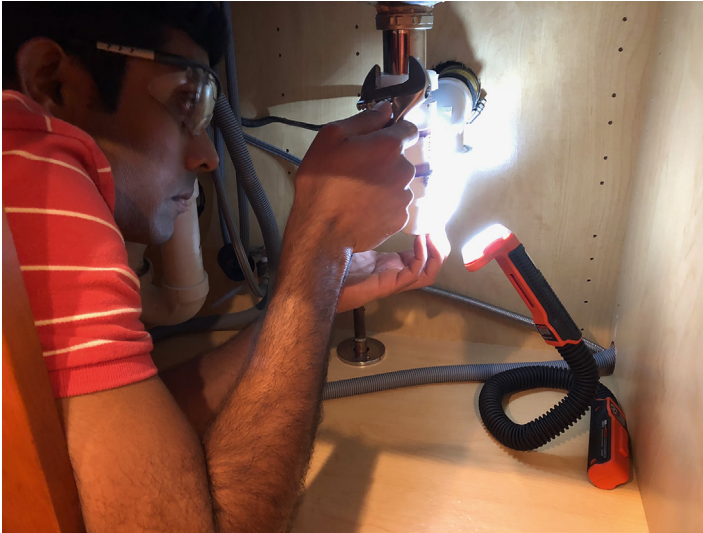


Fig. 6.4
Black+Decker
SnakeLight.

design, but they would not have considered a totally new product like the SnakeLight. Overnight, the SnakeLight had the largest production volume in Black+Decker history, despite being larger, heavier, and more expensive than other flashlights on the market (Segal, 1996).

6.7.4 Hints for Successful Data Gathering

Partner with users. Help the participants understand that you have to deeply know their usage. Probe for specific stories about their usage and how they feel about the product or system.

Be a good listener. Let the users talk and probe with specific questions.

- Do not expect every user to have the same view of the work domain and the work; ask questions about the differences and find ways to combine their views to get the truth.

Be a good detective. Be an effective data ferret or detective; follow leads and discover, deviate from the script, extract, tease out, and collect clues.

Listen to what the user says is needed. Even though it is the job of your team to deduce needs and requirements from the UX research data, users will suggest things they would like to see in the system too. Make raw data notes of any design suggestions and ask for reasons underlying those suggestions. In UX research analysis, these notes will be essential as inputs to user stories as requirements.

Grab the data while you can. Capture the details as they occur; do not wait and try to remember it later.

Be flexible. Be ready to adapt, modify, explore, and branch out.

Avoid interjecting your own views. Do not lead the user or introduce your own perspectives or opinions about what users may need. Follow your leads with questions, not hypotheses, for the user to confirm. For example, consider this user comment: “I want privacy when I am buying tickets.” You may be tempted to say, “You mean, when you are looking for events and buying tickets, you do not want other people in line to know what you are doing?” To which the user might respond, “Yes, that’s what I mean.” A better way to handle the user’s comment here would have been with a follow-up question, such as, “Can you elaborate what you mean by wanting privacy?”

6.7.5 Specific Things to Look For

As you interview and observe to gather data, keep an eye out for these specific kinds of UX research data, each of which is described in future chapters:

- User work roles ([Section 8.2](#))
- User persona information ([Section 8.3](#))
- Inputs to user stories and requirements ([Sections 10.2](#) and [10.3](#))
- Work practice artifacts ([Section 8.5](#))
- Flow of information and artifacts ([Section 8.4](#))
- User tasks ([Sections 9.2](#) and [9.3](#))
- Physical work environment ([Section 8.6](#))
- Information architecture ([Section 8.7](#))

User work roles. As we discussed earlier in this chapter ([Section 6.2.6](#)), a user work role is not a person, but a work assignment defined by the duties, functions, and work activities of a person with a certain job title or job responsibility, such as customer or database administrator. Each work role is annotated with corresponding user class characteristics necessary to carry out that role. More on user work roles in [Section 8.2](#).

User persona information. User personas are UX design targets to guide design to meet the needs of a variety of different kinds of users. User personas help designers focus on specific user characteristics rather than trying to design for all users or even for the average user. Data gathering is a good time to seek out information for building user personas. More on user personas in [Section 8.3](#).

Inputs to user stories and requirements. User stories (narrative descriptions of features desired or needed by users in a specific work role) are used like requirements to drive and inform agile UX design. Inputs to user story writing include statements about the following:

- Things that need to be done by people in various user work roles
- Features that are wanted or needed
- Specific problems or opportunities
- Motivations and potential benefits or payoffs

See [Section 10.2.1](#) about the evolved role of user stories in design requirements.

Work practice artifacts. Artifacts users employ, manipulate, and share as part of their workflow provides a rich source of understanding of the work practice essential to immersion (a form of deep thought and analysis of the problem at hand) by the team. These artifacts are also great conversational props within data gathering as we interview the different roles that use them.

Note how work practice artifacts are created, communicated, and used. For example, guest checks, receipts, and menus are common artifacts associated with the work practice in a restaurant for ordering, preparing, delivering, and charging for food. They provide avenues for discussion given the fact that almost every restaurant uses these artifacts repeatedly.

What things work with this kind of artifact for order taking? What are some breakdowns? How does a person's handwriting impact this part of the work activity? What is the interaction like between the wait staff and the restaurant's guests? What is the interaction like between the wait staff and the kitchen staff?

The UX research data-gathering team must pay close attention to, and take notes on, how work practice artifacts are created, communicated, and used. You can even note ways that you can improve the design of many of the work artifacts. Ask questions such as: What is the purpose of the notes scribbled on the forms? Why are some fields in this form left blank? Why is there a sticky note on this form? Is there a signature required for approval on other kinds of documents? Observers and interviewers should collect as many artifacts as possible during their UX research data-gathering field visits to users.

In addition to physical artifacts, ask about electronic objects that are part of the work practice flow. Think about emails, input forms in software, screenshots, and so on. More on work practice artifacts in [Section 8.5](#).

Flow of information and artifacts. How do information and artifacts flow through the system as work gets done? For example, how is the storage and handling of a customer's car key managed in an auto repair shop? If you have time, it is helpful to capture data flow in a simple sketch (see [Section 8.4](#)) and check your diagram with users to see if they agree.

User tasks. Capture data about user task structure in a simple hierarchical (indented) list of tasks and sub-tasks. Capture data about how to do a few key

tasks, each as a list of a few ordered steps, possibly interspersed with system feedback. More on user tasks in [Sections 9.2 and 9.3](#).

Physical work environment. Sketch a diagram or take photos of the physical workspace layout, workers, and equipment. Make notes about how it impacts work practice. More on physical work environments in [Section 8.6](#).

Information architecture. What information do users access, manipulate, share, output, or archive as part of their work, and how is it structured for storage, retrieval, display, manipulation, and communication? More about information architecture in [Section 8.7](#).

6.7.6 General Things to Look For

Surprises—the interesting and the unusual. In addition to the normal expected things, keep an eye out for the unusual—surprises, interesting episodes, excitement, and joy—to emphasize in the design. By the same token, watch out for negative surprises, disappointments, and breakdowns—things to avoid or fix in the design.

Emotional and social aspects and meaningfulness in work/play practice. In the domain of consumer products, especially personal products, such as a camera, mobile phone, or music player, emotional impact can play a central role in usage and design for the user experience. What evidence is there of joy, aesthetics, and fun in usage? What opportunities are there to design for more of the same? You may have to dig deeper to see evidence of emotional impact within enterprise systems. You may find that customers and users are less likely to mention emotional aspects of their work practice because they think that is about personal feelings. They may think that is inappropriate in the context of technology and functional requirements. So you must try harder to uncover an understanding about emotional and social aspects of work practice. For example, where and when are people having fun? What are they doing when they have fun? What aspects of their work give them a sense of accomplishment or achievement?

With enterprise systems, emotional impact and meaningfulness can also occur in a negative form. Look for ways to fight job boredom. Does anyone hint, even obliquely, that they would like their job to be less boring? What about the work is boring?

Where is there stress and pressure? Where can job stress be relieved with aesthetics and fun? Also be aware of work situations where it would be distracting, dangerous, or otherwise inappropriate to inject fun or surprise.

Meaningfulness in the long term. Meaningfulness is a long-term effect of positive emotional impact usually associated with consumer products. Meaningfulness is manifest as a personal relationship in which users afford the

system or product a presence in their lives. A smartphone is perhaps the most ubiquitous example. Many people become so attached to their smartphones that they cannot be comfortably separated from them.

Meaningfulness goes beyond direct usage. For example, what does a device's brand mean to people who carry it? The Subaru company, for example, prides itself in the values, such as strong corporate social responsibility, it showcases to the community. What does a personal product say about its owner? How about the style and form of the device and how it intersects with the user's personality? Does the user associate a camera, for example, with good times and vacations, taking photos while all the worries are left behind?

Photo ops. Where possible, take photos of the work practice in action to supplement your UX research data. Especially look for opportunities to take photos of the physical work environment and artifacts being used.

Activity-based interaction and the broader ecology. Look for broader user activities that entail sets of related task threads that work together to reach a higher-level goal. Where possible, try to organize UX research data in terms of activities rather than individual tasks. Norman (2005) describes a smartphone as facilitating an activity-based approach to communicating with friends by supporting and integrating the tasks of looking up numbers, dialing, texting, and even emailing.

6.7.7 Capture the UX Research Data

Engage users in the work. One approach to avoid users describing mandated policy instead of actual practice is to ask them to demonstrate what they did the previous day (after ascertaining it was a typical day).

Take raw notes on stream of occurrence. Clients and users are very busy, so you may have only a little time with them. During the visit, especially if the work domain is unfamiliar, you may only have time to take notes. Stream of occurrence note taking involves making notes about whatever is occurring at the time it happens and building on your current set of notes, without taking time to interpret or organize them.

If the domain is familiar and observations happen at a reasonable pace, you may do some sketching to represent workflow and other important characteristics.

Choose the note-taking media that work best for you. In our experience, it is often most effective to take notes by hand, writing in a notebook much as a detective would at a crime scene. Handwriting has the advantage of being the least distracting to the data-gathering process. Furthermore, it is the most flexible with respect to being able to do quick sketches and annotations to notes or any work artifacts.

Some prefer typing in a laptop, but this can distract from your focus on UX research. On the positive side, taking notes on a laptop offers the advantage of immediate electronic sharing of the notes.

Avoid the use of video recording. In almost all cases, we recommend against audio or video recording for UX research data capture in the field. There is usually not time enough to record, and it is often too complicated, somewhat intrusive, makes users reticent about participating freely, and provides little feedback.

6.7.8 Write Effective Raw Data Notes

Writing effective raw UX research data notes is an important technique in the UX technique toolbox (see [Section 2.3](#)). Here is how you do it:

- Be concise in the following ways:
 - Abstract out all but the salient points.
 - Avoid verbatim quotes.
 - Eliminate rambling narration.
 - Paraphrase and synthesize to minimize volume.
- Make your raw data notes as modular as possible by making each work activity note:
 - A simple declarative point
 - Easy to read
 - Understood at a glance
 - Containing just one concept, idea, or fact

Exercise 6.2 UX Research Data Gathering for the Product or System of Your Choice

Goal: Get practice in performing UX research data gathering.

Activities: The best conditions for this exercise are to work as a team and have a real client. If you are working with a team but do not have a real client, divide your team members into users and interviewers and do a role-playing exercise.

Deliverables: At least a few pages of raw UX research data notes, handwritten or typed. Include a few interesting examples (something unexpected or unique) from your notes to share.

Schedule: Assuming a relatively simple domain, we expect you can learn a lot in a couple of hours.

6.7.9 Efficiency in UX Research Data Gathering

With experience, the UX researcher understands the nature of work activity notes and can skip steps in raw UX research data capture, going directly to notes that are final, or near-final, work activity notes (see [Section 7.2](#)).

6.8 ROOTS OF USER RESEARCH

6.8.1 Roots in Activity Theory

We owe a great acknowledgment to those who pioneered, developed, and promoted the concepts and techniques behind user (UX) research. Early foundations go back to activity theory, a theoretical descriptive framework based on the Soviet psychological activity theory adapted by Scandinavian work activity theory researchers in the mid-1980s (Bjerknes, Ehn, & Kyng, 1987; Bødker, 1991; Ehn, 1988).

The goal seems to have been to understand people (users of the systems) as complex human beings and not information processors or system components. This theory is based on learning what motivates people to take actions as influenced by their social and cultural setting. Human activities, especially in the context of interacting with machines and systems, are seen as social actions, considering the individual characteristics and abilities of human operators.

More recent conferences and special issues have been devoted to the topic (Lantz & Gulliksen, 2003). Much of the initial work in this school of thought was directed at the impact of computer-based systems on human labor and democracy within the organizations of the affected workers. This singular focus on human work activities shines through into today's UX research data gathering and analysis.

6.8.2 Roots in Ethnography

A second essential foundation for contextual inquiry is ethnography, an investigative field rooted in anthropology (LeCompte & Preissle, 1993). Anthropologists spend significant amounts of time living with and studying a particular group of humans, usually in social settings of first people's cultures. The goal is to study and document details of their daily lives and existence. In a trend toward design driven by work practice in context, various aspects of ethnography have been adapted into human-computer interaction (HCI) practice as qualitative tools for understanding design requirements.

The characteristics that define ethnography in anthropology are what make it just right for adaptation in HCI, where it takes place in the natural setting of the people being studied. It also involves observation of user activities, listening to what users say, asking questions, and discussing the work with the people who do it; and it is based on the holistic view of understanding behavior in its context.

Lewis et al. (1996) described an ethnographic-based approach to system requirements and design that parallels much of the user research process described here. Rogers and Bellotti (1997) tell how they harnessed ethnography as a practical tool to generate requirements to drive the design process. Blythin, Rouncefield, & Hughes (1997) address the adaptation of ethnography from research to commercial system/product development.

6.8.3 Getting Contextual Studies Into HCI

The foundations for contextual design (design based on work in context) in HCI were laid by researchers at Digital Equipment Corporation (Whiteside & Wixon, 1987; Wixon, 1995; Wixon, Holtzblatt, & Knox, 1990). By 1988, several groups in academia and industry were already reporting on early contextual field studies (Good, 1989) in the United States and the United Kingdom (notably the work of Andrew Monk). Similar trends were also beginning in the software world (Suchman, 1987) and, later, Constantine and Lockwood (1999). Almost a decade later, Wixon and Ramey (1996) produced an edited collection of much more in-depth reports on case studies of real application of contextual studies in the field. Whiteside, Bennett, & Holtzblatt (1988) helped integrate the concept of contextual studies into the UX process. The terms contextual inquiry, contextual design, and contextual studies are early terms for what people are now calling user research, and we call UX research. After this initial work, it was Holtzblatt (in Beyer & Holtzblatt, 1997; Holtzblatt & Beyer, 2016; Holtzblatt, Wendell, & Wood, 2005) who carried the flag for contextual design for many years.

6.9 SEGUE

In this chapter we covered the data-gathering aspects of UX research. In the next chapter we move on to discussing how to analyze these data.